API Integration Guide

American Forests Carbon Dashboard

Spatial Informatics Group

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Purpose and scope

This guide documents how American Forests (AF) developers and data-science staff integrate with the Carbon Model Service API. It consolidates the model-service contents into the handoff structure requested by AF: access and authentication, endpoint and schema reference, integration examples, operational behavior and constraints, and API continuity.

Service overview

The Carbon Model Service is a FastAPI application supporting carbon estimation, carbon-unit conversion, financial pro forma calculations, scenario runs, breakeven solving, model-registry access, filtered GeoJSON, and PDF report generation.

Component	Current implementation
FastAPI application	model_service/main.py - endpoints, startup lifecycle, and report generation
Core modeling	model_service/model.py - FVS/model prediction, carbon, pro forma, scenario, and solve logic
Schemas	model_service/schemas.py - Pydantic request and response models
TPA sweep	model_service/tpa_sweep.py - TPA breakeven sweep/range solver
Geospatial support	model_service/geo.py - filtered GeoJSON generation from the model registry
Configuration sync	model_service/config_sync.py - synchronization of shipped defaults into the model store
Storage abstraction	model_service/store/ - model, configuration, and GeoJSON storage
Python client	aff_dash_client/ - programmatic client and CLI for scenario runs

Modeling pipeline

- Resolve variant, location, PCT, and species inputs
- Use registered FVS-derived model predictions when available
- Convert carbon outputs to protocol-specific carbon units
- Calculate financial pro forma rows and summary metrics
- Optionally solve breakeven acreage or TPA ranges

Environments and base URLs

Environment	API base URL	Dashboard URL
Development	LINK	LINK
Production	LINK	LINK

- Local development default: http://127.0.0.1:8001 when uvicorn is run on port 8001.

- API URL resolution: programmatic clients may pass api_base_url, set CARBON_API_BASE_URL, or use the AFFDashClient default development URL. In production, utils/config.py rejects CARBON_API_BASE_URL values that point to localhost or 127.0.0.1.

Authentication and AF Access

The Carbon Model Service uses machine-to-machine bearer-token authentication. It does not authenticate individual end users, maintain user sessions, or perform an OAuth/token-exchange flow. The American Forests dashboard is issued a static API key and sends that key on every request.

Every API request, except GET /health, must include this HTTP header:

```
None
Authorization: Bearer <AF_API_KEY>
```

The API validates the bearer key before executing the request. If the key is missing, invalid, or revoked, the API returns 401 Unauthorized.

AF client identity

American Forests should be configured as its own client in the API key configuration, for example:

```
None
{
  "af-dashboard": {
    "keys": ["af-key"],
    "role": "client"
  },
  "sig-dashboard": {
    "keys": ["admin-key"],
    "role": "admin"
  }
}
```

The client role allows AF to call the read and compute endpoints needed for integration, such as scenario defaults, scenario runs, bulk scenario runs, registry reads, and related model-service endpoints. Admin-only operations, such as mutating service state or refreshing internal GeoJSON caches, require an admin role and are not expected for AF access.

The service supports multiple keys per client to allow key rotation without downtime:

```
None
{
  "af-dashboard": {
    "keys": ["af-key-1"], ["af-key-2]"
    "role": "client"
  },
  "sig-dashboard": {
    "keys": ["admin-key-1"]
    "role": "admin"
```

```
  }
}
```

During rotation, SIG can add a new key, AF can update its backend to use the new key, and SIG can remove the old key after the transition is complete.

Where keys are configured

API keys are stored in the model store, in GCS, as api_keys.json, alongside registry.json. The key file is shared by the deployed API environment, so key additions, removals, and rotations can be handled without changing application code.

The service can also read the same API key configuration from the CARBON_API_KEYS environment variable. If CARBON_API_KEYS is set, it overrides the api_keys.json file.

Authentication can technically be disabled with CARBON_API_AUTH=off. This is intended only for local development and is refused when ENV=production.

Expected authentication responses

The API returns 200 when the request includes a valid bearer key with sufficient role. It returns 401 Unauthorized when the Authorization header is missing, the bearer token is malformed, the key is unknown, or the key has been revoked. It returns 403 Forbidden when the caller is authenticated but its role is not permitted for the requested endpoint, such as a client-role key calling an admin-only endpoint.

A missing or invalid key returns a response similar to:

```
None
{
  "detail": "Missing or invalid API key"
 }
```

Public health check

The only API route intentionally exempt from bearer authentication is GET /health. This endpoint may be used for service health checks. All other model-service routes should be treated as authenticated API routes.

Implementation references

Authentication is implemented in model_service/auth.py. The FastAPI app applies authentication globally in model_service/main.py using dependencies=[Depends(auth.require_client)].

The bundled Python client attaches the bearer key in aff_dash_client/backends.py and reads CARBON_API_KEY in aff_dash_client/client.py.

The Streamlit dashboard uses the shared authenticated session defined in utils/config.py, where the session sets Authorization: Bearer <key> on outbound requests to the model service.

Endpoint reference

The service currently documents the following endpoint groups. Required and optional fields, data types, units, and response schemas can be checked against the deployed OpenAPI specification at /docs.

Method	Endpoint	Purpose
GET	/health	Service health/status
GET	/carbon/coefficients	Coefficient-based carbon-model parameters
GET	/proforma/presets	Default pro forma assumptions
GET	/variant/presets	Effective variant presets
GET	/species/labels	Species-code labels
GET	/protocol/rules	Carbon protocol rules
GET	/variant/species	Variant-to-species mappings
GET	/geo/variants	Filtered GeoJSON for locations with registered models
POST	/geo/refresh	Rebuild filtered GeoJSON cache
GET	/models/registry	Current model registry
GET	/models/pct-info	PCT levels/retention for a variant/location pair
POST	/carbon/calculate	Calculate carbon metrics
POST	/carbon/units	Convert carbon rows to protocol-specific units
POST	/proforma/compute	Compute pro forma rows and financial summaries
GET	/scenario/defaults	Default scenario inputs for a variant/location pair
POST	/scenario/run	Run full carbon-to-financial scenario pipeline
POST	/scenario/bulk	Evaluate up to 1,000 scenarios
POST	/scenario/solve-tpa	Find TPA ranges where NPV meets a target
POST	/reports/generate	Generate a PDF report with Quarto

Carbon calculation

POST /carbon/calculate uses CarbonInputs. The request fields are variant, location code, survival input, species index, species trees/acre (a list ordered according to the variant species order), and thinning percent, which defaults to PCT0 (No thinning).

Field	Required/default	Units / convention	Description
Variant	Required	Code/string	FVS variant
Loccode	Required	Code/string	Location code

Survival	Required	Value/number	Survival input
SI	Required/variant-dependent	Value/number	Site index / species index field as implemented
Species_tpa	Required	Value/number	Species planting densities
Pct_level	Optional; PCT0 default	PCT code/string	Pre-commercial thinning level

The endpoint first attempts to load matching FVS models by variant, location code, and PCT level. If model output is available it returns FVS-based projections; otherwise it falls back to coefficient-based carbon scores. The response includes carbon_df and model_source, where model_source is FVS or coefficients.

Carbon unit conversion

POST /carbon/units converts calculated carbon rows into protocol-specific carbon-unit results.

The endpoint accepts carbon results, one or more selected carbon protocols, and the applicable protocol rules. It applies the protocol-specific accounting logic to the supplied carbon data rather than rerunning the underlying growth or carbon model.

Field	Purpose
carbon_rows	Carbon calculation records to convert
protocols	Carbon protocols to evaluate
protocol_rules	Optional protocol-rule overrides or ruleset

The response contains the protocol-adjusted carbon-unit rows produced from the supplied carbon data.

This endpoint is useful when an integration already has carbon-model output and needs to evaluate one or more protocols independently.

Pro forma computation

POST /proforma/compute calculates project financial results from carbon-unit data and financial assumptions.

The request supplies the carbon-unit rows used by the financial model together with project acreage, selected protocols, and the financial parameters required for the pro forma calculation.

Financial parameters may include assumptions such as establishment costs, inventory costs, registry fees, verification expenses, issuance fees, carbon prices, discounting assumptions, or protocol-specific overrides, depending on the deployed schema.

The response contains the calculated pro forma rows and summary financial metrics used by the application.

This endpoint is useful when carbon-unit results already exist and only the financial portion of the workflow needs to be recalculated.

Scenario evaluation

POST /scenario/run uses ScenarioRequest and returns ScenarioResponse. Survival input, species index, species tpa, net acres, protocols, and the financial parameters are optional. Thinning level defaults to PCT0. NPV Year defaults to 40. Solve is optional and return_dataframes controls detailed dataframe output. Missing values may be backfilled from default_scenario (variant, loccode).

Field	Units / convention	Notes
Variant	Code/string	Model variant
Loccode	Code/string	Location code
Survival	Value/number	Survival assumption
SI	Value/number	Site index/species index as implemented
Species_tpa	Value/number	Species mix
Pct_level	PCT code/string	PCT selection
Net_acres	Value/number	Project acreage
Protocols	Protocol codes/string	Selected carbon protocols
Financial_params	Value/number (object)	Financial assumptions object
Npv_year	Value/number	NPV evaluation horizon
Solve	Directive/object	Optional solve request
return_dataframes	Boolean	Controls detailed dataframe return

Bulk scenario evaluation

POST /scenario/bulk evaluates multiple scenarios through the same scenario-analysis pipeline.

The request uses BulkScenarioRequest and contains a collection of individual scenario definitions. The service currently supports evaluating up to 1,000 scenarios in a single request.

Each scenario can vary inputs such as variant, location, planting density, survival, site index, acreage, protocol selection, thinning level, or financial assumptions according to the deployed request schema.

The response uses BulkScenarioResponse and returns the results associated with each submitted scenario.

This endpoint should be used when a client needs to evaluate a predefined collection of scenarios efficiently without making a separate HTTP request for each scenario. It differs from /scenario/solve-tpa, which generates and evaluates planting-density values specifically to locate a target financial outcome.

TPA sweep

POST /scenario/solve-tpa uses TpaSweepRequest and supports scalar, species, and per species modes. It accepts a ScenarioRequest plus target npv, comparison operator (>=, <=, or ==), optional grid/species controls, and include_curve. The response includes feasible intervals and optional curve points showing how NPV changes across the TPA sweep.

Mode	Behavior

scalar	Scales the full species mix while preserving species ratios.
species	Varies one selected species while holding others fixed.
per_species	Evaluates each species separately.

Report generation

POST /reports/generate generates a PDF report from supplied scenario and reporting data.

The endpoint uses the report request/data schemas and passes the prepared reporting content into the Quarto-based report-generation workflow.

The request contains the values needed to populate the report, which may include project information, scenario assumptions, carbon results, financial metrics, charts, tables, or other report-specific data according to the deployed schema.

The response provides the generated report output or the information required by the client to retrieve it, depending on the current deployment implementation.

This endpoint should normally be called after the required scenario calculations have been completed and the application has assembled the information that should appear in the final report.

Response schemas and integration guidance

For most integrations, prefer POST /scenario/run rather than separately chaining:

POST /carbon/calculate → POST /carbon/units → POST /proforma/compute

The individual endpoints remain useful when a client needs to rerun only one stage of the calculation pipeline.

Call GET /scenario/defaults before submitting a scenario when an authoritative model and financial defaults are required. Set return_dataframes=true on /scenario/run when detailed proforma_rows, carbon_rows, or cu_rows are needed rather than summary results alone.

Use /scenario/bulk when evaluating a known collection of scenarios and /scenario/solve-tpa when the objective is to search planting-density values for a specified NPV condition.

Major Pydantic schemas include ProformaRequest/Response, CarbonInputs/Response, CarbonUnitsRequest/Response, ReportData/Request, ScenarioRequest/Response/Defaults, BulkScenarioRequest/Response, and TpaSweepRequest/Response.

Example API requests

The following examples demonstrate how to call the primary API endpoints and provide representative request payloads for common integration workflows. Each example shows the expected request structure and illustrates how the fields described in the endpoint reference are passed to the service. These examples

are intended as implementation guidance; available fields, validation requirements, default values, and response schemas should be verified against the deployed FastAPI specification at /docs.

Health check

Returns FastAPI health status.

None
GET {API_BASE_URL}/health

Coefficients, presets, and labels

Returns values from selected config.

None
Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

GET {API_BASE_URL}/carbon/coefficients
GET {API_BASE_URL}/proforma/presets
GET {API_BASE_URL}/variant/presets
GET {API_BASE_URL}/variant/species
GET {API_BASE_URL}/species/labels
GET {API_BASE_URL}/protocol/rules

Registry and site/model discovery

Returns available variants and model information.

None
Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

GET {API_BASE_URL}/models/registry
GET {API_BASE_URL}/models/pct-info
GET {API_BASE_URL}/geo/variants

Rebuild GeoJSON Cache

Rebuild the filtered GeoJSON cache from the current registry.

None
Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

GET {API_BASE_URL}/geo/refresh

Get Default Dictionary

Return a complete defaults dict for a (variant, loccode) pair, ready to feed into /scenario/run. Composes FVSVariant_presets, variant_species, and proforma_presets.

None

Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

GET {API_BASE_URL}/scenario/defaults

Example:
{API_BASE_URL}/scenario/defaults?variant=PN&loccode=609

Carbon workflow example

/carbon/calculate/

Input

None

POST {API_BASE_URL}/carbon/calculate/

Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

```
{
  "variant": "PN",
  "loccode": "609",
  "survival": 0.8,
  "si": 80,
  "species_tpa": [100, 50, 25],
  "pct_level": "PCT0"
}
```

Output

None


```
{
  "carbon_df": [
    {
      "Year": 2026,
      "ABLD_C": 0,
      "BA": 0,
      "DF_TPA": 0,
      "MCuFt": 0,
      "QMD": 0,
      "RC_TPA": 0,
      "SDI": 0,
      "SS_TPA": 0,
      "TCuFt": 0,
      "Tpa": 0,
      "WH_TPA": 0,
```

```
  "Annual_ABLD_C": 0
},
{
  "Year": 2031,
  "ABLD_C": 0.009175323378137,
  "BA": 0.008284538116663038,
  "DF_TPA": 84.99801538952875,
  "MCuFt": 0,
  "QMD": 0.10105053329849975,
  "RC_TPA": 21.25121797678262,
  "SDI": 0,
  "SS_TPA": 0.0024009699455191935,
  "TCuFt": 0,
  "Tpa": 148.7520873424315,
  "WH_TPA": 42.500475793256655,
  "Annual_ABLD_C": 0.009175323378137
},
{
  "Year": 2036,
  "ABLD_C": 0.2024734093903106,
  "BA": 1.5302154958243368,
  "DF_TPA": 81.67801184396754,
  "MCuFt": 0,
  "QMD": 1.3986938396570314,
  "RC_TPA": 20.4513420646172,
  "SDI": 6.112070385671233,
  "SS_TPA": 0,
  "TCuFt": 0,
  "Tpa": 143.014351753925,
  "WH_TPA": 40.888988624220424,
  "Annual_ABLD_C": 0.1932980860121736
},
{
  "Year": 2041,
  "ABLD_C": 1.1943437788285456,
  "BA": 6.539322380565592,
  "DF_TPA": 79.0216615599171,
  "MCuFt": 0,
  "QMD": 3.0095991744791633,
  "RC_TPA": 19.678671865975314,
  "SDI": 19.179411702704424,
  "SS_TPA": 0,
  "TCuFt": 79.94867417896663,
  "Tpa": 129.5821877408358,
  "WH_TPA": 39.525151098208156,
  "Annual_ABLD_C": 0.9918703694382349
},
{
  "Year": 2046,
  "ABLD_C": 7.585989070537652,
  "BA": 30.433903881712734,
  "DF_TPA": 72.39988793267747,
  "MCuFt": 40.732655206693835,
  "QMD": 6.419822773572652,
  "RC_TPA": 16.047333158728613,
```

```
  "SDI": 65.60881586580086,
  "SS_TPA": 4.680083621397451,
  "TCuFt": 374.71494067032967,
  "Tpa": 128.64845668319063,
  "WH_TPA": 35.521155638879506,
  "Annual_ABLD_C": 6.391645291709106
},
{
  "Year": 2051,
  "ABLD_C": 18.159737232704742,
  "BA": 61.642933350369475,
  "DF_TPA": 71.90382733315415,
  "MCuFt": 666.5645108500764,
  "QMD": 9.268994352627276,
  "RC_TPA": 16.042371473012217,
  "SDI": 115.66693718642978,
  "SS_TPA": 4.643205315632095,
  "TCuFt": 923.9714367842466,
  "Tpa": 127.93322451099942,
  "WH_TPA": 35.343821816466445,
  "Annual_ABLD_C": 10.57374816216709
},
{
  "Year": 2056,
  "ABLD_C": 31.354594071413313,
  "BA": 95.61307533758034,
  "DF_TPA": 71.5366069688728,
  "MCuFt": 1360.8383746611676,
  "QMD": 11.640842092408601,
  "RC_TPA": 15.981111796050556,
  "SDI": 164.5069068257447,
  "SS_TPA": 4.530908421922227,
  "TCuFt": 1654.3257554490742,
  "Tpa": 127.21607863485033,
  "WH_TPA": 35.16745548306045,
  "Annual_ABLD_C": 13.19485683870857
},
{
  "Year": 2061,
  "ABLD_C": 46.39446547369874,
  "BA": 130.84725608929233,
  "DF_TPA": 71.17783012773289,
  "MCuFt": 2171.9514034613594,
  "QMD": 13.684200764578934,
  "RC_TPA": 15.913933224862078,
  "SDI": 211.3192370650267,
  "SS_TPA": 4.354907194496157,
  "TCuFt": 2504.002573219659,
  "Tpa": 126.43863666864566,
  "WH_TPA": 34.991964211512304,
  "Annual_ABLD_C": 15.039871402285424
},
{
  "Year": 2066,
  "ABLD_C": 62.00252687780387,
```

```
  "BA": 165.09860037224882,
  "DF_TPA": 70.84449110137356,
  "MCuFt": 3084.653187158738,
  "QMD": 15.437423591842984,
  "RC_TPA": 15.84930031562514,
  "SDI": 254.44329281294347,
  "SS_TPA": 4.278046542026438,
  "TCuFt": 3436.957326503478,
  "Tpa": 125.78915068590753,
  "WH_TPA": 34.817312025321286,
  "Annual_ABLD_C": 15.608061404105129
},
{
  "Year": 2071,
  "ABLD_C": 78.15363089581513,
  "BA": 198.92016062397667,
  "DF_TPA": 70.57945104677293,
  "MCuFt": 4010.3867055014025,
  "QMD": 16.990429272433968,
  "RC_TPA": 15.790311709220436,
  "SDI": 295.3254201482591,
  "SS_TPA": 4.273758128286211,
  "TCuFt": 4536.690278037719,
  "Tpa": 125.2870040263701,
  "WH_TPA": 34.64348223845153,
  "Annual_ABLD_C": 16.15110401801126
},
{
  "Year": 2076,
  "ABLD_C": 94.24418256672968,
  "BA": 230.87725753809912,
  "DF_TPA": 70.32019171429064,
  "MCuFt": 4977.30496554737,
  "QMD": 18.365179029978925,
  "RC_TPA": 15.70324276354776,
  "SDI": 332.51675075688377,
  "SS_TPA": 4.252564969997557,
  "TCuFt": 5636.4416726999,
  "Tpa": 124.6264954390268,
  "WH_TPA": 34.350495721582654,
  "Annual_ABLD_C": 16.09055167091455
},
{
  "Year": 2081,
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  "BA": 259.7853213483543,
  "DF_TPA": 70.75031669677442,
  "MCuFt": 5886.237509781943,
  "QMD": 19.57439224309593,
  "RC_TPA": 15.255115239737894,
  "SDI": 364.98154572608706,
  "SS_TPA": 3.8972664645079647,
  "TCuFt": 6609.695756065764,
  "Tpa": 123.59945342728389,
  "WH_TPA": 33.69675863708866,
```

```
  "Annual_ABLD_C": 15.682885094235104
},
{
  "Year": 2086,
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  "BA": 285.8009101356233,
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  "MCuFt": 6785.924059746541,
  "QMD": 20.652094724785563,
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  "SDI": 393.21590945588014,
  "SS_TPA": 3.1587662132817442,
  "TCuFt": 7523.389050415666,
  "Tpa": 122.3123251059131,
  "WH_TPA": 33.03284723693175,
  "Annual_ABLD_C": 15.178147946091968
},
{
  "Year": 2091,
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  "BA": 310.07166480474626,
  "DF_TPA": 72.49147493379903,
  "MCuFt": 7514.146229602024,
  "QMD": 21.63406458160258,
  "RC_TPA": 13.535953641259841,
  "SDI": 418.88909474065053,
  "SS_TPA": 2.2956990734333544,
  "TCuFt": 8280.475349870218,
  "Tpa": 121.01801359750516,
  "WH_TPA": 32.694878439068276,
  "Annual_ABLD_C": 14.599309098534974
},
{
  "Year": 2096,
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  "BA": 335.8159126996703,
  "DF_TPA": 73.01835533421844,
  "MCuFt": 8253.547513632599,
  "QMD": 22.535342192656056,
  "RC_TPA": 13.119312031703487,
  "SDI": 446.4534251936218,
  "SS_TPA": 1.9645193998011798,
  "TCuFt": 9140.457581945344,
  "Tpa": 120.90356044401162,
  "WH_TPA": 32.80137660784611,
  "Annual_ABLD_C": 13.70128380250253
},
{
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  "BA": 360.509746063059,
  "DF_TPA": 73.01408860444648,
  "MCuFt": 9279.632209247464,
  "QMD": 23.372497233662596,
  "RC_TPA": 12.826173485277732,
```

```
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  "TCuFt": 10184.525755187478,
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  "WH_TPA": 32.993044564392186,
  "Annual_ABLD_C": 13.4109881121272
},
{
  "Year": 2106,
  "ABLD_C": 179.90482408341168,
  "BA": 385.2391964964723,
  "DF_TPA": 72.60943877595557,
  "MCuFt": 10504.813543589356,
  "QMD": 24.16010428684838,
  "RC_TPA": 12.755892436504048,
  "SDI": 498.25598312566774,
  "SS_TPA": 1.9790706367840336,
  "TCuFt": 11456.224777505344,
  "Tpa": 120.663699737529,
  "WH_TPA": 33.31929504053122,
  "Annual_ABLD_C": 13.08802746319023
},
{
  "Year": 2111,
  "ABLD_C": 192.78034878851832,
  "BA": 409.49375016576266,
  "DF_TPA": 71.98406031344715,
  "MCuFt": 11884.943750428141,
  "QMD": 24.900657123659244,
  "RC_TPA": 12.853864987187976,
  "SDI": 523.2534847330438,
  "SS_TPA": 2.123383084150504,
  "TCuFt": 12864.36973192229,
  "Tpa": 120.727418091023,
  "WH_TPA": 33.76611265890625,
  "Annual_ABLD_C": 12.875524705106642
},
{
  "Year": 2116,
  "ABLD_C": 205.32065644440138,
  "BA": 431.01053572506953,
  "DF_TPA": 70.85654346137372,
  "MCuFt": 13080.965531573529,
  "QMD": 25.60187203975041,
  "RC_TPA": 12.994446451945489,
  "SDI": 544.7558226037851,
  "SS_TPA": 2.225130618476655,
  "TCuFt": 14188.069081882779,
  "Tpa": 120.21827176073091,
  "WH_TPA": 34.142147622683616,
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},
{
  "Year": 2121,
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      "MCuFt": 14279.927292422686,
      "QMD": 26.26992918413769,
      "RC_TPA": 13.209220848989519,
      "SDI": 563.0947202279517,
      "SS_TPA": 2.355433307432076,
      "TCuFt": 15432.409804774388,
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      "WH_TPA": 34.32257549060791,
      "Annual_ABLD_C": 11.915937593570106
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    {
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      "MCuFt": 15329.235087403542,
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      "RC_TPA": 13.313608926661734,
      "SDI": 578.004992725452,
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      "TCuFt": 16563.789085111956,
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      "WH_TPA": 34.38559995222116,
      "Annual_ABLD_C": 11.07758899883865
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  "model_source": "fvs"
}
```


/carbon/units/

Input

None

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POST {API_BASE_URL}/carbon/units/

Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

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    {
      "Year": 2036,
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{
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  "Year": 2046,
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{
  "Year": 2051,
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{
  "Year": 2056,
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{
  "Year": 2061,
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{
  "Year": 2066,
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{
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{
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  "protocols": ["ACR"]
}
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Output

None


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   },
   {
    "Year": 2126,
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  ]
}
```


/proforma/compute/

Input

None

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POST {API_BASE_URL}/proforma/compute/

Content-Type: application/json
Authorization: Bearer {AF_API_KEY}
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  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2099,
  "CU_ac": 7.868494945147177,
  "Project_acres": 100,
  "CU": 786.8494945147177,
  "CUs_Sold": 0,
  "CU_Credit_Price": 104.02850937628791,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
```

```
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2100,
  "CU_ac": 7.868494945147267,
  "Project_acres": 100,
  "CU": 786.8494945147268,
  "CUs_Sold": 0,
  "CU_Credit_Price": 106.10907956381368,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2101,
  "CU_ac": 7.868494945147267,
  "Project_acres": 100,
  "CU": 786.8494945147268,
  "CUs_Sold": 0,
  "CU_Credit_Price": 108.23126115508997,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 38250,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 46750,
  "Net_Revenue": -46750,
  "Protocol": "ACR"
},
{
  "Year": 2102,
  "CU_ac": 7.679007473203001,
  "Project_acres": 100,
  "CU": 767.9007473203001,
  "CUs_Sold": 3915.298725379207,
  "CU_Credit_Price": 110.39588637819176,
  "Total_Revenue": 432232.873223642,
  "Validation_and_Verification": 50000,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 783.0597450758414,
  "Planting_Cost": 0,
  "Total_Costs": 59283.05974507584,
  "Net_Revenue": 372949.8134785662,
  "Protocol": "ACR"
},
```

```
{
  "Year": 2103,
  "CU_ac": 7.679007473203001,
  "Project_acres": 100,
  "CU": 767.9007473203001,
  "CUs_Sold": 0,
  "CU_Credit_Price": 112.60380410575559,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2104,
  "CU_ac": 7.679007473203001,
  "Project_acres": 100,
  "CU": 767.9007473203001,
  "CUs_Sold": 0,
  "CU_Credit_Price": 114.85588018787072,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2105,
  "CU_ac": 7.67900747320291,
  "Project_acres": 100,
  "CU": 767.900747320291,
  "CUs_Sold": 0,
  "CU_Credit_Price": 117.15299779162814,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2106,
  "CU_ac": 7.679007473203001,
  "Project_acres": 100,
```

```
  "CU": 767.9007473203001,
  "CUs_Sold": 0,
  "CU_Credit_Price": 119.49605774746071,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 38250,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 46750,
  "Net_Revenue": -46750,
  "Protocol": "ACR"
},
{
  "Year": 2107,
  "CU_ac": 7.554327854980147,
  "Project_acres": 100,
  "CU": 755.4327854980147,
  "CUs_Sold": 3827.035774779206,
  "CU_Credit_Price": 121.8859789024099,
  "Total_Revenue": 466462.0017035062,
  "Validation_and_Verification": 50000,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 765.4071549558412,
  "Planting_Cost": 0,
  "Total_Costs": 59265.40715495584,
  "Net_Revenue": 407196.5945485504,
  "Protocol": "ACR"
},
{
  "Year": 2108,
  "CU_ac": 7.554327854980238,
  "Project_acres": 100,
  "CU": 755.4327854980238,
  "CUs_Sold": 0,
  "CU_Credit_Price": 124.3236984804581,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2109,
  "CU_ac": 7.554327854980147,
  "Project_acres": 100,
  "CU": 755.4327854980147,
  "CUs_Sold": 0,
  "CU_Credit_Price": 126.81017245006727,
  "Total_Revenue": 0,
```

```
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2110,
  "CU_ac": 7.554327854980147,
  "Project_acres": 100,
  "CU": 755.4327854980147,
  "CUs_Sold": 0,
  "CU_Credit_Price": 129.34637589906862,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2111,
  "CU_ac": 7.554327854980147,
  "Project_acres": 100,
  "CU": 755.4327854980147,
  "CUs_Sold": 0,
  "CU_Credit_Price": 131.93330341705,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 38250,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 46750,
  "Net_Revenue": -46750,
  "Protocol": "ACR"
},
{
  "Year": 2112,
  "CU_ac": 7.357649307859719,
  "Project_acres": 100,
  "CU": 735.764930785972,
  "CUs_Sold": 3757.49607277804,
  "CU_Credit_Price": 134.571969485391,
  "Total_Revenue": 505653.6468473629,
  "Validation_and_Verification": 50000,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 751.499214555608,
```

```
  "Planting_Cost": 0,
  "Total_Costs": 59251.49921455561,
  "Net_Revenue": 446402.1476328073,
  "Protocol": "ACR"
},
{
  "Year": 2113,
  "CU_ac": 7.357649307859719,
  "Project_acres": 100,
  "CU": 735.764930785972,
  "CUs_Sold": 0,
  "CU_Credit_Price": 137.26340887509883,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2114,
  "CU_ac": 7.357649307859719,
  "Project_acres": 100,
  "CU": 735.764930785972,
  "CUs_Sold": 0,
  "CU_Credit_Price": 140.0086770526008,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2115,
  "CU_ac": 7.357649307859629,
  "Project_acres": 100,
  "CU": 735.7649307859629,
  "CUs_Sold": 0,
  "CU_Credit_Price": 142.80885059365283,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
```

```
},
{
  "Year": 2116,
  "CU_ac": 7.357649307859719,
  "Project_acres": 100,
  "CU": 735.764930785972,
  "CUs_Sold": 0,
  "CU_Credit_Price": 145.6650276055259,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 38250,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 46750,
  "Net_Revenue": -46750,
  "Protocol": "ACR"
},
{
  "Year": 2117,
  "CU_ac": 6.991318904899435,
  "Project_acres": 100,
  "CU": 699.1318904899435,
  "CUs_Sold": 3642.1916136338223,
  "CU_Credit_Price": 148.5783281576364,
  "Total_Revenue": 541150.7407834773,
  "Validation_and_Verification": 50000,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 728.4383227267645,
  "Planting_Cost": 0,
  "Total_Costs": 59228.43832272677,
  "Net_Revenue": 481922.30246075056,
  "Protocol": "ACR"
},
{
  "Year": 2118,
  "CU_ac": 6.991318904899527,
  "Project_acres": 100,
  "CU": 699.1318904899526,
  "CUs_Sold": 0,
  "CU_Credit_Price": 151.54989472078913,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2119,
  "CU_ac": 6.991318904899435,
```

```
  "Project_acres": 100,
  "CU": 699.1318904899435,
  "CUs_Sold": 0,
  "CU_Credit_Price": 154.5808926152049,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2120,
  "CU_ac": 6.991318904899435,
  "Project_acres": 100,
  "CU": 699.1318904899435,
  "CUs_Sold": 0,
  "CU_Credit_Price": 157.672510467509,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2121,
  "CU_ac": 6.991318904899435,
  "Project_acres": 100,
  "CU": 699.1318904899435,
  "CUs_Sold": 0,
  "CU_Credit_Price": 160.82596067685918,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 38250,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 46750,
  "Net_Revenue": -46750,
  "Protocol": "ACR"
},
{
  "Year": 2122,
  "CU_ac": 6.499443017398607,
  "Project_acres": 100,
  "CU": 649.9443017398607,
  "CUs_Sold": 3446.471863699644,
  "CU_Credit_Price": 164.04247989039638,
```

```
  "Total_Revenue": 565367.7913937658,
  "Validation_and_Verification": 50000,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 689.2943727399288,
  "Planting_Cost": 0,
  "Total_Costs": 59189.294372739925,
  "Net_Revenue": 506178.4970210259,
  "Protocol": "ACR"
},
{
  "Year": 2123,
  "CU_ac": 6.499443017398607,
  "Project_acres": 100,
  "CU": 649.9443017398607,
  "CUs_Sold": 0,
  "CU_Credit_Price": 167.3233294882043,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2124,
  "CU_ac": 6.499443017398699,
  "Project_acres": 100,
  "CU": 649.9443017398698,
  "CUs_Sold": 0,
  "CU_Credit_Price": 170.66979607796839,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
  "Issuance_Fees": 0,
  "Planting_Cost": 0,
  "Total_Costs": 8500,
  "Net_Revenue": -8500,
  "Protocol": "ACR"
},
{
  "Year": 2125,
  "CU_ac": 6.499443017398517,
  "Project_acres": 100,
  "CU": 649.9443017398517,
  "CUs_Sold": 0,
  "CU_Credit_Price": 174.08319199952777,
  "Total_Revenue": 0,
  "Validation_and_Verification": 0,
  "Survey_Cost": 0,
  "Registry_Fees": 8500,
```

```
      "Issuance_Fees": 0,
      "Planting_Cost": 0,
      "Total_Costs": 8500,
      "Net_Revenue": -8500,
      "Protocol": "ACR"
    },
    {
      "Year": 2126,
      "CU_ac": 6.499443017398607,
      "Project_acres": 100,
      "CU": 649.9443017398607,
      "CUs_Sold": 0,
      "CU_Credit_Price": 177.56485583951834,
      "Total_Revenue": 0,
      "Validation_and_Verification": 0,
      "Survey_Cost": 38250,
      "Registry_Fees": 8500,
      "Issuance_Fees": 0,
      "Planting_Cost": 0,
      "Total_Costs": 46750,
      "Net_Revenue": -46750,
      "Protocol": "ACR"
    }
  ],
  "summaries": [
    {
      "Protocol": "ACR",
      "total_net": 2801161.3673609225,
      "npv_yr": -334832.75812711474,
      "npv_year": 20,
      "npv_per_acre": -3348.327581271147
    }
  ]
}
```


Full carbon scenario workflow

/scenario/run/

Input

None

```
POST {API_BASE_URL}/scenario/run/

Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

{
        "variant":"PN",
        "loccode":"609",
        "survival":0.8,
        "si":80,
```

```
            "species_tpa":[100,50,25],
            "pct_level":"PCT0",
            "net_acres":1000,
            "protocols":["ACR"],
            "npv_year":40,
            "return_dataframes":false
}
```

Output

```
None
{
  "inputs": {
    "variant": "PN",
    "loccode": "609",
    "survival": 0.8,
    "si": 80,
    "species_tpa": [
      100,
      50,
      25
    ],
    "species_codes": [
      "DF",
      "WH",
      "RC",
      "SS"
    ],
    "pct_level": "PCT0",
    "net_acres": 1000,
    "protocols": [
      "ACR"
    ],
    "npv_year": 40,
    "financial_params": {
      "ACR": {
        "num_plots": 250,
        "cost_per_cfi_plot": 150,
        "price_per_ert_initial": 25,
        "credit_price_increase": 0.02,
        "registry_fees": 500,
        "validation_cost": 45000,
        "verification_cost": 25000,
        "issuance_fee_per_ert": 0.15,
        "anticipated_inflation": 0.02,
        "discount_rate": 0.06,
        "planting_cost": 1000,
        "protocol_overrides": {
          "ACR": {
            "cost_per_cfi_plot": 150,
            "registry_fees": 8500,
            "issuance_fee_per_ert": 0.2,
            "validation_cost": 15000,
            "verification_cost": 50000,
```

```
          "planting_cost": 1000
        },
        "CAR": {
          "cost_per_cfi_plot": 150,
          "registry_fees": 1500,
          "issuance_fee_per_ert": 0.2,
          "validation_cost": 15000,
          "verification_cost": 50000,
          "planting_cost": 1000
        },
        "VERRA": {
          "cost_per_cfi_plot": 150,
          "registry_fees": 11750,
          "issuance_fee_per_ert": 0.23,
          "validation_cost": 15000,
          "verification_cost": 50000,
          "planting_cost": 1000
        },
        "GS": {
          "cost_per_cfi_plot": 150,
          "registry_fees": 7500,
          "issuance_fee_per_ert": 0.25,
          "validation_cost": 15000,
          "verification_cost": 50000,
          "planting_cost": 1000
        },
        "ISO": {
          "cost_per_cfi_plot": 150,
          "registry_fees": 0,
          "issuance_fee_per_ert": 0,
          "validation_cost": 0,
          "verification_cost": 0,
          "planting_cost": 1000
        }
       }
      }
    }
  },
  "summaries": [
    {
      "Protocol": "ACR",
      "net_acres": 1000,
      "total_net": 2953305.822343843,
      "npv_yr": 893493.886382376,
      "npv_year": 40,
      "npv_per_acre": 893.493886382376
    }
  ],
  "model_source": "fvs",
  "proforma_rows": null,
  "carbon_rows": null,
  "cu_rows": null
}
```

Solver workflow example

/scenario/solve-tpa/

Input

None

```
POST {API_BASE_URL}/scenario/solve-tpa/

Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

{
  "variant": "PN",
  "loccode": "609",
  "survival": 85,
  "si": 80,
  "species_tpa": [100, 50, 25, 0],
  "pct_level": "PCT0",
  "net_acres": 100,
  "protocols": ["ACR"],
  "financial_params": {
    "ACR": {
      "net_acres": 100,
      "num_plots": 250,
      "cost_per_cfi_plot": 150,
      "price_per_ert_initial": 25,
      "credit_price_increase": 0.02,
      "registry_fees": 8500,
      "validation_cost": 15000,
      "verification_cost": 50000,
      "issuance_fee_per_ert": 0.2,
      "anticipated_inflation": 0.02,
      "discount_rate": 0.06,
      "planting_cost": 1000
    }
  },
  "npv_year": 20,
  "mode": "scalar",
  "op": ">=",
  "target_npv": 0,
  "grid": {
    "lo_factor": 0.25,
    "hi_factor": 4,
    "max_value": 450,
    "steps": 25
  },
  "include_curve": true
}
```

Output

None

```
{
  "mode": "scalar",
```

```
"op": ">=",
"target_npv": 0,
"metric": "total_npv",
"base_species_tpa": [
  100,
  50,
  25,
  0
],
"species_codes": [
  "DF",
  "WH",
  "RC",
  "SS"
],
"results": [
  {
    "species_index": null,
    "species_code": null,
    "variable": "k",
    "range": null,
    "intervals": [],
    "curve": [
      {
        "x": 0.25,
        "npv": -349770.2207899362,
        "species_tpa": [
          25,
          12.5,
          6.25,
          0
        ]
      },
      {
        "x": 0.40625,
        "npv": -346346.69871649175,
        "species_tpa": [
          40.625,
          20.3125,
          10.15625,
          0
        ]
      },
      {
        "x": 0.5625,
        "npv": -343102.36434944585,
        "species_tpa": [
          56.25,
          28.125,
          14.0625,
          0
        ]
      },
      {
        "x": 0.71875,
```

```
  "npv": -340020.9173372192,
  "species_tpa": [
    71.875,
    35.9375,
    17.96875,
    0
  ]
},
{
  "x": 0.875,
  "npv": -337086.0573282326,
  "species_tpa": [
    87.5,
    43.75,
    21.875,
    0
  ]
},
{
  "x": 1.03125,
  "npv": -334281.4839709067,
  "species_tpa": [
    103.125,
    51.5625,
    25.78125,
    0
  ]
},
{
  "x": 1.1875,
  "npv": -331590.89691366226,
  "species_tpa": [
    118.75,
    59.375,
    29.6875,
    0
  ]
},
{
  "x": 1.34375,
  "npv": -328997.99580492,
  "species_tpa": [
    134.375,
    67.1875,
    33.59375,
    0
  ]
},
{
  "x": 1.5,
  "npv": -326486.4802931007,
  "species_tpa": [
    150,
    75,
    37.5,
```

```
      0
    ]
  },
  {
    "x": 1.65625,
    "npv": -324040.05002662505,
    "species_tpa": [
      165.625,
      82.8125,
      41.40625,
      0
    ]
  },
  {
    "x": 1.8125,
    "npv": -321642.4046539138,
    "species_tpa": [
      181.25,
      90.625,
      45.3125,
      0
    ]
  },
  {
    "x": 1.96875,
    "npv": -319277.2438233878,
    "species_tpa": [
      196.875,
      98.4375,
      49.21875,
      0
    ]
  },
  {
    "x": 2.125,
    "npv": -316928.2671834675,
    "species_tpa": [
      212.5,
      106.25,
      53.125,
      0
    ]
  },
  {
    "x": 2.28125,
    "npv": -314579.1743825741,
    "species_tpa": [
      228.125,
      114.0625,
      57.03125,
      0
    ]
  },
  {
    "x": 2.4375,
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```
  "npv": -312213.66506912786,
  "species_tpa": [
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    121.875,
    60.9375,
    0
  ]
},
{
  "x": 2.59375,
  "npv": -309815.43889154977,
  "species_tpa": [
    259.375,
    129.6875,
    64.84375,
    0
  ]
},
{
  "x": 2.75,
  "npv": -307368.19549826055,
  "species_tpa": [
    275,
    137.5,
    68.75,
    0
  ]
},
{
  "x": 2.90625,
  "npv": -304855.63453768083,
  "species_tpa": [
    290.625,
    145.3125,
    72.65625,
    0
  ]
},
{
  "x": 3.0625,
  "npv": -302261.45565823145,
  "species_tpa": [
    306.25,
    153.125,
    76.5625,
    0
  ]
},
{
  "x": 3.21875,
  "npv": -299569.3585083331,
  "species_tpa": [
    321.875,
    160.9375,
    80.46875,
```

```
        0
      ]
    },
    {
      "x": 3.375,
      "npv": -296763.04273640655,
      "species_tpa": [
        337.5,
        168.75,
        84.375,
        0
      ]
    },
    {
      "x": 3.53125,
      "npv": -293826.20799087256,
      "species_tpa": [
        353.125,
        176.5625,
        88.28125,
        0
      ]
    },
    {
      "x": 3.6875,
      "npv": -290742.55392015184,
      "species_tpa": [
        368.75,
        184.375,
        92.1875,
        0
      ]
    },
    {
      "x": 3.84375,
      "npv": -287495.780172665,
      "species_tpa": [
        384.375,
        192.1875,
        96.09375,
        0
      ]
    },
    {
      "x": 4,
      "npv": -284069.58639683295,
      "species_tpa": [
        400,
        200,
        100,
        0
      ]
    }
  ]
}
```

```
  ]
}
```

Report workflow example

/reports/generate/

Input

None

```
POST {API_BASE_URL}/reports/generate/

Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

{
  "data": {
    "planting_design": [
      {
        "column1": "Reforestation Strategy",
        "column2": "Mixed Species Planting"
      },
      {
        "column1": "Variant",
        "column2": "PN"
      },
      {
        "column1": "Location Name",
        "column2": "PN 609"
      },
      {
        "column1": "Location Code",
        "column2": "609"
      },
      {
        "column1": "Area, acres",
        "column2": 100
      },
      {
        "column1": "Survival Rate, %",
        "column2": 85
      },
      {
        "column1": "Site Index, ft",
        "column2": 80
      },
      {
        "column1": "Included Protocols",
        "column2": "ACR"
      },
      {
        "column1": "PCT Level",
        "column2": "PCT0"
```

```
  },
  {
    "column1": "PCT Retention, %",
    "column2": ""
  }
],
"species_mix": [
  {
    "column1": "Douglas-fir",
    "column2": 100
  },
  {
    "column1": "western hemlock",
    "column2": 50
  },
  {
    "column1": "red cedar",
    "column2": 25
  },
  {
    "column1": "Sitka spruce",
    "column2": 0
  }
],
"financial_options1": [
  {
    "column1": "Net Acres",
    "column2": 100
  },
  {
    "column1": "Number of CFI Plots",
    "column2": 250
  },
  {
    "column1": "Cost per CFI Plot",
    "column2": 150
  },
  {
    "column1": "Initial Credit Price",
    "column2": 25
  },
  {
    "column1": "Credit Price Increase",
    "column2": "2%"
  },
  {
    "column1": "Anticipated Inflation",
    "column2": "2%"
  },
  {
    "column1": "Discount Rate",
    "column2": "6%"
  },
  {
    "column1": "NPV Horizon",
```

```
    "column2": 20
  }
],
"financial_options2": [
  {
    "column1": "Protocol",
    "column2": "ACR"
  },
  {
    "column1": "Registry Fees",
    "column2": 8500
  },
  {
    "column1": "Validation Cost",
    "column2": 15000
  },
  {
    "column1": "Verification Cost",
    "column2": 50000
  },
  {
    "column1": "Issuance Fee per ERT",
    "column2": 0.2
  },
  {
    "column1": "Planting Cost per Acre",
    "column2": 1000
  }
],
"carbon": [
  {
    "Year": 2027,
    "Protocol": "ACR",
    "CU": 0.00538334573242054,
    "Cumulative_CU": 0.00538334573242054,
    "CO2e": 0.00538334573242054,
    "Annual CO2e per acre": 0.00538334573242054,
    "Annual COer per acre": 0.00538334573242054,
    "Annual CO2e": 0.538334573242054,
    "TotalRevenue": 13.45836433105135,
    "TotalCosts": 123500.10766691465,
    "NetRevenue": -123486.6493025836
  },
  {
    "Year": 2028,
    "Protocol": "ACR",
    "CU": 0.00538334573242054,
    "Cumulative_CU": 0.01076669146484108,
    "CO2e": 0.01076669146484108,
    "Annual CO2e per acre": 0.00538334573242054,
    "Annual COer per acre": 0.00538334573242054,
    "Annual CO2e": 0.538334573242054,
    "TotalRevenue": 0,
    "TotalCosts": 8500,
    "NetRevenue": -8500
```

```
},
{
  "Year": 2029,
  "Protocol": "ACR",
  "CU": 0.005383345732420543,
  "Cumulative_CU": 0.016150037197261623,
  "CO2e": 0.016150037197261623,
  "Annual CO2e per acre": 0.005383345732420543,
  "Annual COer per acre": 0.005383345732420543,
  "Annual CO2e": 0.5383345732420543,
  "TotalRevenue": 0,
  "TotalCosts": 8500,
  "NetRevenue": -8500
},
{
  "Year": 2030,
  "Protocol": "ACR",
  "CU": 0.005383345732420539,
  "Cumulative_CU": 0.021533382929682162,
  "CO2e": 0.021533382929682162,
  "Annual CO2e per acre": 0.005383345732420539,
  "Annual COer per acre": 0.005383345732420539,
  "Annual CO2e": 0.5383345732420539,
  "TotalRevenue": 0,
  "TotalCosts": 46000,
  "NetRevenue": -46000
},
{
  "Year": 2031,
  "Protocol": "ACR",
  "CU": 0.005383345732420539,
  "Cumulative_CU": 0.0269167286621027,
  "CO2e": 0.0269167286621027,
  "Annual CO2e per acre": 0.005383345732420539,
  "Annual COer per acre": 0.005383345732420539,
  "Annual CO2e": 0.5383345732420539,
  "TotalRevenue": 71.35900627276929,
  "TotalCosts": 58500.53833457324,
  "NetRevenue": -58429.179328300474
},
{
  "Year": 2032,
  "Protocol": "ACR",
  "CU": 0.11341185302506251,
  "Cumulative_CU": 0.14032858168716522,
  "CO2e": 0.14032858168716522,
  "Annual CO2e per acre": 0.11341185302506251,
  "Annual COer per acre": 0.11341185302506251,
  "Annual CO2e": 11.341185302506251,
  "TotalRevenue": 0,
  "TotalCosts": 8500,
  "NetRevenue": -8500
},
{
  "Year": 2033,
```

```
  "Protocol": "ACR",
  "CU": 0.11341185302506249,
  "Cumulative_CU": 0.2537404347122277,
  "CO2e": 0.2537404347122277,
  "Annual CO2e per acre": 0.11341185302506249,
  "Annual COer per acre": 0.11341185302506249,
  "Annual CO2e": 11.34118530250625,
  "TotalRevenue": 0,
  "TotalCosts": 8500,
  "NetRevenue": -8500
},
{
  "Year": 2034,
  "Protocol": "ACR",
  "CU": 0.11341185302506256,
  "Cumulative_CU": 0.36715228773729027,
  "CO2e": 0.36715228773729027,
  "Annual CO2e per acre": 0.11341185302506256,
  "Annual COer per acre": 0.11341185302506256,
  "Annual CO2e": 11.341185302506257,
  "TotalRevenue": 0,
  "TotalCosts": 8500,
  "NetRevenue": -8500
},
{
  "Year": 2035,
  "Protocol": "ACR",
  "CU": 0.11341185302506247,
  "Cumulative_CU": 0.48056414076235274,
  "CO2e": 0.48056414076235274,
  "Annual CO2e per acre": 0.11341185302506247,
  "Annual COer per acre": 0.11341185302506247,
  "Annual CO2e": 11.341185302506247,
  "TotalRevenue": 0,
  "TotalCosts": 46000,
  "NetRevenue": -46000
},
{
  "Year": 2036,
  "Protocol": "ACR",
  "CU": 0.11341185302506247,
  "Cumulative_CU": 0.5939759937874152,
  "CO2e": 0.5939759937874152,
  "Annual CO2e per acre": 0.11341185302506247,
  "Annual COer per acre": 0.11341185302506247,
  "Annual CO2e": 11.341185302506247,
  "TotalRevenue": 1568.094024085653,
  "TotalCosts": 58511.34118530251,
  "NetRevenue": -56943.24716121686
},
{
  "Year": 2041,
  "Protocol": "ACR",
  "CU": 0.5819501831568008,
  "Cumulative_CU": 3.503726909571421,
```

```
  "CO2e": 3.503726909571421,
  "Annual CO2e per acre": 0.5819501831568008,
  "Annual COer per acre": 0.5819501831568008,
  "Annual CO2e": 58.19501831568008,
  "TotalRevenue": 9000,
  "TotalCosts": 58600,
  "NetRevenue": -49600
},
{
  "Year": 2046,
  "Protocol": "ACR",
  "CU": 3.7501061255515653,
  "Cumulative_CU": 22.25425753732925,
  "CO2e": 22.25425753732925,
  "Annual CO2e per acre": 3.7501061255515653,
  "Annual COer per acre": 3.7501061255515653,
  "Annual CO2e": 375.01061255515654,
  "TotalRevenue": 90000,
  "TotalCosts": 59000,
  "NetRevenue": 31000
},
{
  "Year": 2051,
  "Protocol": "ACR",
  "CU": 6.2038295217066715,
  "Cumulative_CU": 53.27340514586261,
  "CO2e": 53.27340514586261,
  "Annual CO2e per acre": 6.2038295217066715,
  "Annual COer per acre": 6.2038295217066715,
  "Annual CO2e": 620.3829521706672,
  "TotalRevenue": 200000,
  "TotalCosts": 60000,
  "NetRevenue": 140000
},
{
  "Year": 2056,
  "Protocol": "ACR",
  "CU": 7.741686404407085,
  "Cumulative_CU": 91.98183716789809,
  "CO2e": 91.98183716789809,
  "Annual CO2e per acre": 7.741686404407085,
  "Annual COer per acre": 7.741686404407085,
  "Annual CO2e": 774.1686404407085,
  "TotalRevenue": 300000,
  "TotalCosts": 61000,
  "NetRevenue": 239000
},
{
  "Year": 2061,
  "Protocol": "ACR",
  "CU": 8.82419334914889,
  "Cumulative_CU": 136.10280391364254,
  "CO2e": 136.10280391364254,
  "Annual CO2e per acre": 8.82419334914889,
  "Annual COer per acre": 8.82419334914889,
```

```
  "Annual CO2e": 882.419334914889,
  "TotalRevenue": 400000,
  "TotalCosts": 62000,
  "NetRevenue": 338000
},
{
  "Year": 2066,
  "Protocol": "ACR",
  "CU": 9.157561787016538,
  "Cumulative_CU": 181.89061284872523,
  "CO2e": 181.89061284872523,
  "Annual CO2e per acre": 9.157561787016538,
  "Annual COer per acre": 9.157561787016538,
  "Annual CO2e": 915.7561787016538,
  "TotalRevenue": 450000,
  "TotalCosts": 63000,
  "NetRevenue": 387000
},
{
  "Year": 2071,
  "Protocol": "ACR",
  "CU": 9.476175749447567,
  "Cumulative_CU": 229.27149159596308,
  "CO2e": 229.27149159596308,
  "Annual CO2e per acre": 9.476175749447567,
  "Annual COer per acre": 9.476175749447567,
  "Annual CO2e": 947.6175749447567,
  "TotalRevenue": 500000,
  "TotalCosts": 64000,
  "NetRevenue": 436000
},
{
  "Year": 2076,
  "Protocol": "ACR",
  "CU": 9.440648476359002,
  "Cumulative_CU": 276.4747339777581,
  "CO2e": 276.4747339777581,
  "Annual CO2e per acre": 9.440648476359002,
  "Annual COer per acre": 9.440648476359002,
  "Annual CO2e": 944.0648476359002,
  "TotalRevenue": 520000,
  "TotalCosts": 65000,
  "NetRevenue": 455000
},
{
  "Year": 2081,
  "Protocol": "ACR",
  "CU": 9.201462342489595,
  "Cumulative_CU": 322.4820456902061,
  "CO2e": 322.4820456902061,
  "Annual CO2e per acre": 9.201462342489595,
  "Annual COer per acre": 9.201462342489595,
  "Annual CO2e": 920.1462342489595,
  "TotalRevenue": 530000,
  "TotalCosts": 66000,
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```
  "NetRevenue": 464000
},
{
  "Year": 2086,
  "Protocol": "ACR",
  "CU": 8.905322962931042,
  "Cumulative_CU": 367.0086605048613,
  "CO2e": 367.0086605048613,
  "Annual CO2e per acre": 8.905322962931042,
  "Annual COer per acre": 8.905322962931042,
  "Annual CO2e": 890.5322962931042,
  "TotalRevenue": 540000,
  "TotalCosts": 67000,
  "NetRevenue": 473000
},
{
  "Year": 2091,
  "Protocol": "ACR",
  "CU": 8.565706634292429,
  "Cumulative_CU": 409.83719367632344,
  "CO2e": 409.83719367632344,
  "Annual CO2e per acre": 8.565706634292429,
  "Annual COer per acre": 8.565706634292429,
  "Annual CO2e": 856.5706634292429,
  "TotalRevenue": 550000,
  "TotalCosts": 68000,
  "NetRevenue": 482000
},
{
  "Year": 2096,
  "Protocol": "ACR",
  "CU": 8.038817232604288,
  "Cumulative_CU": 450.0312798393449,
  "CO2e": 450.0312798393449,
  "Annual CO2e per acre": 8.038817232604288,
  "Annual COer per acre": 8.038817232604288,
  "Annual CO2e": 803.8817232604288,
  "TotalRevenue": 560000,
  "TotalCosts": 69000,
  "NetRevenue": 491000
},
{
  "Year": 2101,
  "Protocol": "ACR",
  "CU": 7.868494945147267,
  "Cumulative_CU": 489.3737545650812,
  "CO2e": 489.3737545650812,
  "Annual CO2e per acre": 7.868494945147267,
  "Annual COer per acre": 7.868494945147267,
  "Annual CO2e": 786.8494945147267,
  "TotalRevenue": 570000,
  "TotalCosts": 70000,
  "NetRevenue": 500000
},
{
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```
  "Year": 2106,
  "Protocol": "ACR",
  "CU": 7.679007473203001,
  "Cumulative_CU": 527.7687919310962,
  "CO2e": 527.7687919310962,
  "Annual CO2e per acre": 7.679007473203001,
  "Annual COer per acre": 7.679007473203001,
  "Annual CO2e": 767.9007473203001,
  "TotalRevenue": 580000,
  "TotalCosts": 71000,
  "NetRevenue": 509000
},
{
  "Year": 2111,
  "Protocol": "ACR",
  "CU": 7.554327854980147,
  "Cumulative_CU": 565.5404312059969,
  "CO2e": 565.5404312059969,
  "Annual CO2e per acre": 7.554327854980147,
  "Annual COer per acre": 7.554327854980147,
  "Annual CO2e": 755.4327854980147,
  "TotalRevenue": 590000,
  "TotalCosts": 72000,
  "NetRevenue": 518000
},
{
  "Year": 2116,
  "Protocol": "ACR",
  "CU": 7.357649307859719,
  "Cumulative_CU": 602.3286777452955,
  "CO2e": 602.3286777452955,
  "Annual CO2e per acre": 7.357649307859719,
  "Annual COer per acre": 7.357649307859719,
  "Annual CO2e": 735.7649307859719,
  "TotalRevenue": 600000,
  "TotalCosts": 73000,
  "NetRevenue": 527000
},
{
  "Year": 2121,
  "Protocol": "ACR",
  "CU": 6.991318904899435,
  "Cumulative_CU": 637.2852722697927,
  "CO2e": 637.2852722697927,
  "Annual CO2e per acre": 6.991318904899435,
  "Annual COer per acre": 6.991318904899435,
  "Annual CO2e": 699.1318904899435,
  "TotalRevenue": 610000,
  "TotalCosts": 74000,
  "NetRevenue": 536000
},
{
  "Year": 2126,
  "Protocol": "ACR",
  "CU": 6.499443017398607,
```

```
      "Cumulative_CU": 669.7824873567857,
      "CO2e": 669.7824873567857,
      "Annual CO2e per acre": 6.499443017398607,
      "Annual COer per acre": 6.499443017398607,
      "Annual CO2e": 649.9443017398607,
      "TotalRevenue": 620000,
      "TotalCosts": 75000,
      "NetRevenue": 545000
    }
   ],
   "selected_variant": "PN"
  }
}
```


Output

- Formatted link for downloading generated PDF report.

Bulk Scenario Example

Evaluate up to MAX_BULK_BATCH_SIZE scenarios in one request.

/scenario/bulk/

Input

None

```
POST {API_BASE_URL}/scenario/bulk/

Content-Type: application/json
Authorization: Bearer {AF_API_KEY}

{
  "scenarios": [
    {
      "variant": "PN",
      "loccode": "609",
      "survival": 0.8,
      "si": 80,
      "species_tpa": [100, 50, 25, 0],
      "pct_level": "PCT0",
      "net_acres": 1000,
      "protocols": ["ACR"],
      "financial_params": null,
      "npv_year": 40,
      "solve": null,
      "return_dataframes": false
    }
  ]
}
```

Output

None


```
{
  "results": [
    {
      "inputs": {
        "variant": "PN",
        "loccode": "609",
        "survival": 0.8,
        "si": 80,
        "species_tpa": [
          100,
          50,
          25,
          0
        ],
        "species_codes": [
          "DF",
          "WH",
          "RC",
          "SS"
        ],
        "pct_level": "PCT0",
        "net_acres": 1000,
        "protocols": [
          "ACR"
        ],
        "npv_year": 40,
        "financial_params": {
          "ACR": {
            "num_plots": 250,
            "cost_per_cfi_plot": 150,
            "price_per_ert_initial": 25,
            "credit_price_increase": 0.02,
            "registry_fees": 500,
            "validation_cost": 45000,
            "verification_cost": 25000,
            "issuance_fee_per_ert": 0.15,
            "anticipated_inflation": 0.02,
            "discount_rate": 0.06,
            "planting_cost": 1000,
            "protocol_overrides": {
              "ACR": {
                "cost_per_cfi_plot": 150,
                "registry_fees": 8500,
                "issuance_fee_per_ert": 0.2,
                "validation_cost": 15000,
                "verification_cost": 50000,
                "planting_cost": 1000
              },
              "CAR": {
                "cost_per_cfi_plot": 150,
                "registry_fees": 1500,
                "issuance_fee_per_ert": 0.2,
                "validation_cost": 15000,
                "verification_cost": 50000,
```

```
            "planting_cost": 1000
          },
          "VERRA": {
            "cost_per_cfi_plot": 150,
            "registry_fees": 11750,
            "issuance_fee_per_ert": 0.23,
            "validation_cost": 15000,
            "verification_cost": 50000,
            "planting_cost": 1000
          },
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            "issuance_fee_per_ert": 0,
            "validation_cost": 0,
            "verification_cost": 0,
            "planting_cost": 1000
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        "npv_per_acre": 893.493886382376
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    "proforma_rows": null,
    "carbon_rows": null,
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 ],
 "errors": []
}
```


Operational behavior and constraints

Characteristic	Documented behavior

Bulk scenario limit	/scenario/bulk evaluates up to 1,000 scenarios
Model loading	Model files are loaded lazily on first use
Model caching	Loaded models are cached after first use
GeoJSON cache	Built during startup; /geo/refresh rebuilds it
Cloud Run timeout	Existing deployment runbook uses 3,600 seconds
Cloud Run resources	Existing runbook uses 2 GiB RAM / 2 CPU / max 3 instances

Startup and caching behavior

On startup, the service loads the model registry, synchronizes default configuration from conf/base, and builds the cached filtered GeoJSON used by the dashboard map. Model files are loaded lazily on first use rather than loading all models at startup. refresh_geojson() rebuilds the cached filtered GeoJSON from the configured model store.

Bulk and TPA-solve client behavior: HTTPBackend uses client timeouts of at least 300 seconds for these longer-running calls. Production Cloud Run is configured with a 3,600-second request timeout. Bulk execution supports process or thread workers depending on BULK_WORKER_MODE; the current deployment workflow sets BULK_WORKERS=4 and BULK_WORKER_MODE=process.

Known failure modes to document

- 401/403 when the caller is not authenticated or lacks Cloud Run invocation permission.
- 422 request validation errors for malformed or incomplete Pydantic request bodies.
- Verified repository behavior includes FastAPI validation errors for malformed schemas, HTTP 400 for unsupported variants in report generation, and HTTP 404 for missing defaults or registry entries on applicable endpoints. Long-running/bulk clients use extended timeouts, and the production Cloud Run API timeout is configured to 3,600 seconds. Authentication-specific 401/403 behavior should be finalized after the production service is made private.
- Backend service accounts cannot read the configured model store/GCS assets.
- Quarto or report assets are unavailable in the runtime image, causing report generation to fail.
- Long-running solver/report requests reach the configured Cloud Run request timeout.
- A cold instance may have higher first-request latency because models are loaded lazily.

API continuity

Item	Handoff value
Development API URL	LINK
Production API URL	LINK
Hosting owner	Spatial Informatics Group (SIG)
Maintenance/support period	Refer to Hosting and Maintenance Agreement (B1)
Breaking-change notice process	Contact Becca Winterringer and/or Elijah Dalton
Depracation timeline	At the end of Hosting and Maintenance Agreement (B1) timeline

Outage/escalation contact	Becca Winterringer and/or Elijah Dalton

Dashboard integration

For full details on how the API is integrated with the dashboard, please see the Application Deployment and Architecture Guide.

The Streamlit dashboard calls the model service through get_api_base_url() and requests. Site Selection uses GeoJSON and registry-aware variant availability. Planting Design calls carbon, carbon-unit, pro forma, and report workflows.The Solver calls scenario endpoints for breakeven acreage, sensitivity grids, and TPA sweeps. Model Management updates the model registry and associated metadata outside production.

Programmatic integration is also supported through AFFDashClient in aff_dash_client/client.py. HTTPBackend calls /scenario/defaults, /scenario/run, /scenario/bulk, and /scenario/solve-tpa. LocalBackend can import model_service directly and run against a local models directory without starting FastAPI. High-level methods include defaults(), run(), solve_for_tnr(), solve_for_npv(), and solve_tpa_range().

Production integration checks

- Set CARBON_API_BASE_URL to the production API base URL.
- Ensure the dashboard runtime identity can invoke the private Cloud Run API.
- Confirm the API runtime identity can read models/configuration and any required cloud services.
- Verify registry.json and filtered GeoJSON expose the intended production model set.
- Verify ENV=production hides Model Management.